Ovi Paul

Phone: +1 (832) 271-4266 | Email: opaul@uh.edu | Location: Houston, TX, USA
LinkedIn: linkedin.com/in/ovipaul | GitHub: github.com/ovipaul | Google Scholar: Ovi Paul | Website: ovipaul.github.io

Education

University of Houston - *Ph.D. in Geosensing System Engineering & Science* 2023 – Present
Supervisor - Dr. Craig L. Glennie *Houston, Texas, USA*
Thesis: Cross-view image geo-localization.

Independent University, Bangladesh - *Bachelor of Science in Computer Science and Engineering* 2015 - 2019
Minor in Economics *Dhaka, Bangladesh*
Supervisor - Dr. AKM Mahbubur Rahman
Thesis: Deep Learning for Land Use and Land Cover Classification in Dhaka, Bangladesh.

Skills

Programming Languages: Python, C++, Matlab, Java

Web Development: HTML, CSS, JavaScript, REST APIs

Frameworks: PyTorch, Scikit-learn, Flask, Streamlit

Libraries: OpenCV, NumPy, Pandas, Matplotlib, GDAL, PDAL

GIS Tools: QGIS, ArcGIS Pro, MicroStation, TerraScan

Databases: MySQL, PostgreSQL

Environment & Tools: Anaconda, LaTeX

DevOps & Deployment: Linux, Git, Bash, Docker, CI/CD, SSH, SCP, Tmux

Experience

National Center for Airborne Laser Mapping (NCALM) Aug 2023 – Present
Graduate Research Assistant *Houston, Texas, USA*

- Designed a cross-view geo-localization framework using Siamese and Transformer networks with realistic FOV sampling and monocular depth estimation to use relative depth for better geometric consistency, achieving 87% in F-1 score.
- Built a deep learning NER system using fine-tuned BERT and LLaMA models for geospatial inference from tweets, improving location extraction F1-score from 0.74 to 0.88 and mapping geographic coordinates directly from text.
- Segmented bathymetric LiDAR into surface and benthic layers and performed automated underwater correction by modeling laser pulse refraction using deep learning (3D CNNs and graph networks). Reduced mean bias from 8 cm to 2 cm and RMSE from 47 cm to 16 cm compared with previous best methods.

AinoviQ Jan 2022 – Jun 2023
Machine Learning Engineer *Dhaka, Bangladesh*

- Optimized state-of-the-art GANs for realistic cloth-on-body synthesis using DensePose, keypoints, and body segmentation.
- Structured end-to-end workflows for data preprocessing, multi-stage model training, and real-time inference.
- Reduced processing time of multi-layer deep learning pipeline to 3 seconds through architectural and runtime optimizations.
- Conducted systematic testing and fine-tuning to enhance synthesis quality and model efficiency.
- Researched cutting-edge generative models to inform design decisions and improve visual realism.

Center for Computational & Data Sciences (CCDS) May 2022 – April 2023
Research Assistant *Dhaka, Bangladesh*

- Developed BD-Sat, a high-resolution (2.22m) LULC dataset for Dhaka to improve model accuracy in dense urban settings.
- Built attention-based deep learning models for urban scene classification and disaster damage assessment.
- Applied FCN-8 on RGB satellite imagery for semantic segmentation without relying on multispectral inputs.
- Designed automated pipelines for data preprocessing, training, and evaluation to streamline experimentation.
- Contributed to peer-reviewed publications and maintained technical documentation for reproducibility.

Artificial Intelligence and Cybernetics Lab (AGenCy Lab) Feb 2020 - Dec 2021
Research Assistant *Dhaka, Bangladesh*

- Worked on deep learning model optimization.
- Labeled and processed satellite imagery.
- Wrote papers and maintained reproducible codebases.

Research Projects

Deep Learning for Land Use and Land Cover Classification Feb 2020 - Dec 2021
Supervisor: Dr. AKM Mahbubur Rahman (Independent University Bangladesh) *AGenCy Lab*

- Developed semantic segmentation models using DeepLabV3+ with ResNet-101 to label aerial and satellite images for land classification in Bangladesh.
- Automated the labeling process to create ground truth data and performed Land Use and Land Cover analysis.
- Achieved classification accuracy ranging from 0.82 to 0.96 through experiment and optimization.

Deep Learning for Urban Building Categorization Aug 2020 - Jul 2021
Supervisor: Prof. Ryosuke Shibasaki (University of Tokyo) *H-U Tokyo Lab and DnD Lab*
Supervisor: Dr. Moinul Islam Zaber (University of Dhaka)

- Designed and implemented a deep learning framework to classify urban areas into four socioeconomic categories.
- Worked with Prof. Ryosuke Shibasaki and Dr. Moinul Islam Zaber on developing a model for cities in the global south.
- Achieved 91.57 percent average classification accuracy on city segmentation tasks.

Teaching Experience

University of Houston 2025 - Present
Teaching Assistant *Houston, TX, USA*
CIVE 3397: Probability and Statistics for Engineers Semester: Fall 2025
Instructor: Prof. Craig L. Glennie

Light of Hope 2021 - 2022
Coding Instructor *Dhaka, Bangladesh*

- Taught programming basics, data analysis, game and web development.
- Delivered lessons and practical exercises.

Independent University Bangladesh 2016 - 2019
Teaching Assistant *Dhaka, Bangladesh*

CSE 471: Introduction to High Performance Computing (GPU programming) Semester: Fall 2019
Instructor: Dr. AKM Mahbubur Rahman

CSC317: Numerical Method Semester: Summer 2019, Fall 2019
Instructor: Dr. AKM Mahbubur Rahman

CSE 211: Algorithms Semester: Spring 2018
Instructor: Prof. Ali Shihab Sabbir

CSC204: Data Structure Semester: Spring 2017, Summer 2017, Fall 2017
Instructor: Dr. Mahabub Murshed

CSC101: Introduction to Programming Semester: Summer 2016, Fall 2016
Instructor: Md Rakibul Alam

Achievements

Intra IUB Programming Contest **IEEE**
Winner *2018*
ICPC Asia Dhaka Regional Contest **ACM**
Regional Qualifier *2018*
Makers' Mania 2018 (QuadCopter) **IUB**
Second Runner Up *2018*
ICPC Bangladesh IUBAT NCPC **ACM**
Regional Qualifier *2018*
IEEE Day Programming Contest **IEEE**
Winner *2017*

Conference Presentations

American Society for Photogrammetry and Remote Sensing 2024 Gulf South Region Annual Conference **ASPRS**
Bathymetric LiDAR Depth Correction Using Semantic Segmentation and a PointCNN Deep Learning Model *April 2024*

International Conference on Bangla Speech and Language Processing **ICBSLP**
Image Pre-processing on NumtaDB for Bengali Handwritten Digit Recognition *Sep 2018*

Publications

- **O. Paul**, N. Ekhtari, and C. L. Glennie (2025). Automated depth correction of bathymetric LiDAR point clouds using PointCNN semantic segmentation. Frontiers in Remote Sensing, 6:1521446.
- **O. Paul**, A. B. S. Nayem, A. Sarker, et al. (2025). BOLM high resolution land use and land cover dataset and benchmark results for the rapidly developing City of Dhaka Bangladesh. Nature Scientific Reports, 15, 37689.
- S. K. Paul, **O. Paul**, M. Nicolescu, M. Nicolescu. ROS-Pose: Simplified object detection and planar pose estimation for rapid robotics application development. Elsevier, Software Impacts, Vol. 19, 2024, 100624.
- Q. Cheng, AKM M. Rahman, A. Sarker, ABS Nayem, **O. Paul**, A.A. Ali, Md. A. Amin, R. Shibasaki, M. Zaber, "A Deep-learning framework customized for novel categorization method to classify cities of the developing world," Sensors Journal, MDPI, 2021.
- F.F. Niloy, Arif, ABS Nayem, A. Sarker, **O. Paul**, Md.A. Amin, A.A. Ali, AKM M. Rahman, "A Novel Disaster Image Data-set and Characteristics Analysis using Attention Model," 25th ICPR, Milano, Italy, 2021.
- Q. Cheng, AKM M. Rahman, A. Sarker, ABS Nayem, **O. Paul**, A.A. Ali, Md. A. Amin, R. Shibasaki, M. Zaber, "Deep-learning coupled with novel categorization method to classify the urban environment of the developing world," SIGML 2021, Zurich, 2021.
- ABS Nayem, A. Sarker, **O. Paul**, A. Ali, Md. A. Amin, AKM M. Rahman, "LULC Segmentation of RGB Satellite Image Using FCN-8," SLAAI-ICAI, Sri Lanka, 2019.
- **O. Paul**, "Image Pre-processing on NumtaDB for Bengali Handwritten Digit Recognition," ICBSLP, Sylhet, 2018.

Projects

BathyNet: Built a deep learning system in PyTorch using PointCNN to classify seafloor features from 3D LAS point-cloud data, improving accuracy in bathymetric and underwater terrain mapping. [link]

Text2Map: Developed an NLP and geospatial analysis pipeline that uses BERT and Hugging Face Transformers to extract and geocode location entities from text, producing interactive spatial visualisations with GeoPandas and Folium. [link]

Adaptive Retracker: A developing project that leverages machine learning to analyze waveform altimetry data from diverse land types, classify surface characteristics, and perform adaptive retracking to improve elevation accuracy across complex terrains. [link]

Under-the-Sea: Created an interactive web platform in Streamlit to simulate sea-level rise impacts using elevation and census data with PostGIS, GeoPandas, and Rasterio for large-scale spatial computations. [link]

Professional Service

Journal

International Journal of Applied Earth Observation and Geoinformation — **Elsevier**
Reviewer — *2025 - Present*

Journal of Open Source Software — **JOSS**
Reviewer — *2025 - Present*

ReScience C — **ReScience C**
Reviewer — *2021 - Present*

Conference

International Conference on Artificial Intelligence, Computer, Data Sciences and Applications — **Antalya, Turkiye**
Reviewer — *August 7, 2025*

5th International Conference on Electrical, Computer and Energy Technologies — **Paris, France**
Reviewer — *July 2, 2025*

4th International Conference on Electrical, Computer, Communications and Mechatronics Engineering — **Male, Maldives**
Reviewer — *November 3, 2024*

International Conference on Electrical and Computer Engineering Researches — **Gaborone, Botswana**
Reviewer — *December 4, 2024*

26th International Conference on Pattern Recognition — **Montreal, Canada**
Technical Committee Reviewer — *August 21, 2022*

Relevant Courses

University of Houston: Photogrammetry, Deep Learning, LiDAR Systems, Remote Sensing

Independent University Bangladesh: Computer Vision, Artificial Intelligence, Machine Learning, Data Mining, Software Engineering